

Review – Complex Numbers and Quadratic Functions

Simplify.

1. $\sqrt{169} - \sqrt{-289}$ 2. $\frac{13}{\sqrt{6}}$ 3. $\frac{\sqrt{18}}{\sqrt{3}}$ 4. $\sqrt{\frac{-4}{3}}$ 5. $(3 - i) - (5 + 2i)$

6. $(10 + \sqrt{-25}) + (3 - \sqrt{-16}) - (7 + \sqrt{-9})$ 7. $(-6 - 5i)(3 + 4i)$

8. $(\sqrt{40} - \sqrt{-98}) - (\sqrt{90} + \sqrt{-32})$ 9. $(5 + \sqrt{-2})(5 - \sqrt{-2})$

10. $(2 - 3i) \div (-4i)$ 11. $(2 + \sqrt{-16}) \div (4 - \sqrt{-1})$ 12. $(3 + 5i) \div (1 - i)$

SOLVE

13. $2x^2 = 32$

14. $1/3x^2 = 4$

Solve by factoring:

15. $x^2 - 9x = -18$

16. $3x^2 = 3x + 60$

Solve by completing the square:

17. $x^2 - 8x + 3 = 0$

18. $2x^2 + 8x = -18$

Solve using the quadratic formula:

19. $3x^2 = 2x - 7$

20. $x^2 = -5x + 1$

Solve any way you choose:

21. $75x^3 - 50x^2 = 0$

22. $x^2 + 49 = 0$

23. $12x^2 - 36x = 0$

24. $(5x - 2)^2 = 18$

25. $x^2 + 11x + 24 = 0$

26. $5x^2 = 2x - 1$

27. $x(x - 2) = 15$

28. $x^2 + 6x - 10 = 0$